Question 1

Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse-stuffing TDM with no synchronization bits. Answer the following questions: (12 marks)

a. What is the size of the frame?

b. What is the frame rate?

c. What is the duration of the frame?

d. What is the data rate of the link?

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Question 2

Draw the graph of the signal for the bit stream of 1(five consecutive) then a zero, then 1(five consecutive) knowing that the most recent non-zero signal level for bit 1 was negative for AMI, for pseudoternary most recent bit 0 was negative : (9 marks)

a. NRZ-L

b. Bipolar AMI

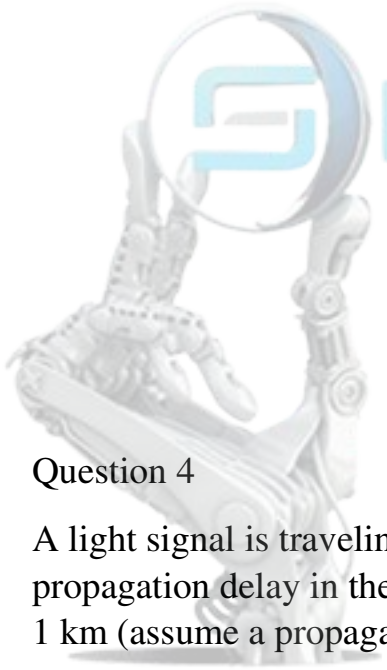
c. Psuedoternary

Question 3

A voice signal with a frequency range from 300 Hz to 3000 Hz is sampled at a rate of 7000 samples per second.

a. If SNR=30dB, find the number of quantization levels (L).

b. What is the recommended data rate?



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Question 4

A light signal is traveling through a fiber-optic cable. What is the propagation delay in the signal if the length of the cable is 10 m, 100 m, and 1 km (assume a propagation speed of 2 10^8 m)?

Question 5

How many bits per baud can be sent in each of the following cases if the signal constellation has the given number of points? (11 marks)

a. 2 b. 4 c. 16 d. 1024



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